Topological Data Analysis of Ryuichi Sakamoto's *hibari*: A Persistent Homology Pipeline for Structure-Preserving Music Generation

Minju Kim
Department of Mathematics
POSTECH (Pohang University of Science and Technology)
Pohang, Republic of Korea

***Abstract*—We present a computational pipeline that extracts the topological structure of a musical composition via persistent homology and re-synthesizes new musical pieces that preserve that structure. Applied to Ryuichi Sakamoto's *hibari* (from the 2009 album *out of noise*), the pipeline (i) constructs a weighted note-network from the MIDI quantization, (ii) computes H_1 persistent homology under four distance functions (frequency, Tonnetz, voice-leading, DFT), (iii) builds an overlap matrix recording cycle activations across the timeline, and (iv) generates new music via either probabilistic sampling (Algorithm 1) or deep learning (Algorithm 2; FC / LSTM / Transformer).**

Over $N = 20$ independent trials, the Tonnetz distance reduces pitch Jensen–Shannon divergence from 0.0753 ± 0.0033 (frequency baseline) to 0.0398 ± 0.0031, a 47% improvement (Welch $t = 35.1$, $p < 10^{-20}$, Cohen's $d = 11.1$). Continuous overlap with per-cycle thresholding adds a further 47.5% gain ($p < 0.001$), and soft activation input to the FC model achieves JS $= 0.0004$, less than 0.1% of the theoretical maximum $\ln 2$. The simplest FC model outperforms LSTM and Transformer on *hibari*, while this ranking reverses on the chromatic *solari*—confirming that a composition's aesthetic character determines the optimal model architecture.

We further extend the pipeline to topology-preserving *variation*: note redistribution under Tonnetz matching constraints (DTW $+61.4\%$), temporal reordering, and harmonic constraints, producing music that is topologically similar yet melodically distinct from the original.

***Index Terms*—Persistent homology, Topological data analysis, Music generation, Tonnetz, Jensen–Shannon divergence, Multi-label classification, Vietoris–Rips complex, Overlap matrix.**

I. Introduction

Topological Data Analysis (TDA) offers a principled way to extract geometric structure from data that is robust to noise and to uniform rescaling of the metric [1], [2]. In this work, we apply TDA to a specific musical composition—Ryuichi Sakamoto's *hibari*—and ask whether the topological invariants (specifically, the first persistent homology group H_1) can serve as a structural seed for generating new, stylistically faithful music.

A. Research Questions

1) How can the topological structure of music be defined mathematically via Vietoris–Rips H_1 persistence?
2) Can new music be generated that *preserves* this structure, and how do we measure preservation quantitatively?
3) Does the choice of distance function (frequency, Tonnetz, voice-leading, DFT) significantly affect generation quality?

B. Why hibari?

hibari is drawn from Sakamoto's experimental album *out of noise* (2009). It offers: (i) adequate scale ($T = 1{,}088$ eighth-note timepoints, $N = 23$ distinct notes, $C = 17$ chords); (ii) a balance of simplicity (7 pitch classes, C major) and complexity (two instruments with coprime period structure, §IV-G0b); (iii) an aesthetic that privileges spatial note placement over temporal causality, directly relevant to our model-selection findings (§IV-E).

C. Contributions

1) A four-stage pipeline (preprocessing $\to$ PH $\to$ overlap matrix $\to$ generation) with two generators and four distance functions.
2) Systematic $N = 20$ statistical comparisons establishing Tonnetz as optimal for diatonic compositions.
3) Continuous overlap matrices with per-cycle thresholds (§VII-G), yielding an additional 47.5% JS gain.
4) Topology-preserving musical variation via Tonnetz-matching note redistribution, temporal reordering, and harmonic constraints (§VII).
5) Cross-composition validation on *solari*, Bach, and Ravel, confirming that the composition's character determines the optimal distance function and model architecture (§VII-B).

II. Mathematical Background

A. Vietoris–Rips Complex

Definition 1. *Given a finite metric space (X, d) and $\varepsilon > 0$, the* Vietoris–Rips complex $\mathrm{VR}_\varepsilon(X)$ *is the simplicial complex* $\mathrm{VR}_\varepsilon(X) = \{\sigma \subseteq X \mid \forall x_i, x_j \in \sigma,\ d(x_i, x_j) \le \varepsilon\}$.

As ε increases, the nested family $\mathrm{VR}_0 \subseteq \mathrm{VR}_{\varepsilon_1} \subseteq \cdots$ forms a *filtration*. In this paper, X consists of the 23 distinct notes of *hibari*.

B. Persistent Homology

The n-th homology group $H_n(K) = \ker \partial_n / \operatorname{im} \partial_{n+1}$ captures n-dimensional "holes." We focus on H_1, whose generators correspond to topological cycles. *Persistent homology* tracks each cycle's birth b and death d across the filtration; $\mathrm{pers} = d - b$ measures stability. Cycles with large persistence represent robust structural features.

We primarily use the pHcol algorithm from prior work [6], [7] with cycle representatives, supplemented by Ripser [5] for speed-critical steps.

C. Musical Distance Functions

We compare four distance functions between notes $n_i = (p_i, d_i)$:

a) Frequency distance (baseline): $d_{\mathrm{freq}}(n_i, n_j) = 1/w(n_i, n_j)$, where w is the adjacency frequency in the chord sequence.

b) Tonnetz distance: Pitch classes are embedded on a hexagonal lattice with edges at $\pm 3, \pm 4, \pm 7$ semitones (minor 3rd, major 3rd, perfect 5th). The distance is the shortest graph path, pre-computed into a 12×12 table (range 0–4) [3].

c) Voice-leading distance: $d_V(p_1, p_2) = |p_1 - p_2|$, the unsigned semitone difference [4].

d) DFT distance: Each pitch class p is encoded as a 12-dimensional indicator vector $\mathbf{e}_p \in \{0,1\}^{12}$. Its discrete Fourier transform yields magnitude coefficients $|F_k|$ for $k = 0, \ldots, 5$ (the first six components suffice by symmetry). The DFT distance is then the L_2 distance between these magnitude vectors: $d_{\mathrm{DFT}}(p_1, p_2) = \sqrt{\sum_{k=0}^{5} (|F_k(p_1)| - |F_k(p_2)|)^2}$. The maximum value is $\sqrt{2}$ (attained by maximally dissimilar pitch classes such as C vs. F♯) [4].

e) Hybrid distance and note-level extensions: All music-theoretic distances are combined with frequency via $d_{\mathrm{hybrid}} = \alpha\, d_{\mathrm{freq}} + (1 - \alpha)\, d_{\mathrm{music}}$, where $\alpha \in [0, 1]$. Each note-level distance further adds an octave term $w_o |o_1 - o_2|$ and a duration term $w_d |d_1 - d_2| / \max(d_1, d_2)$:

$$d(n_1, n_2) = d_{\mathrm{base}} + w_o |o_1 - o_2| + w_d \frac{|d_1 - d_2|}{\max(d_1, d_2)}, \quad (1)$$

with $w_o = 0.3$ (optimized via grid search, §IV-B) and $w_d = 0.3$.

D. Weight Decomposition and Rate Parameter

The note-network weights are decomposed into three components reflecting *hibari*'s two-instrument structure:

- **Intra weights** W_{intra}: within-instrument sequential transitions.
- **Inter weights** W_{inter}: cross-instrument transitions at lag $\ell \in \{1, 2, 3, 4\}$, combined with empirically chosen decayed weights that concentrate on nearby time-lags:

$$W_{\mathrm{inter}} = \sum_{\ell=1}^{4} \lambda_\ell W_{\mathrm{inter}}^{(\ell)}, \quad (2)$$

  with $(\lambda_1, \ldots, \lambda_4) = (0.60, 0.30, 0.08, 0.02)$. The coefficients were set heuristically so that lag 1 dominates while distant lags contribute marginally (sum $= 1.0$). This decayed weighting reduced JS by 69.6% ($0.0398 \to 0.0121$).
- **Simul weights** W_{simul}: simultaneous note co-occurrences.

The *timeflow weight* $W_{\mathrm{tf}}(r) = W_{\mathrm{intra}} + r \cdot W_{\mathrm{inter}}$ is swept over $r \in [0, 1.5]$.

E. Overlap Matrix

Definition 2 (Activation matrix)**.** *For cycle c with vertex set $V(c)$ and timepoint t,*

$$A[t, c] = \begin{cases} 1 & \textit{if at least one note } n \in V(c) \textit{ is played at time } t, \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 3 (Binary overlap matrix)**.** *Let scale_c denote the minimum number of consecutive active timepoints required for cycle c to be considered "sustained" (determined from each cycle's birth/death range in the barcode). Let $R(c)$ be the union of all maximal contiguous intervals $[t_0, t_1]$ in which $A[t, c] = 1$ and $t_1 - t_0 + 1 \geq \mathrm{scale}_c$. Then:*

$$O[t, c] = \begin{cases} 1 & \textit{if } t \in R(c), \\ 0 & \textit{otherwise.} \end{cases}$$

Definition 4 (Continuous overlap)**.**

$$O_{\mathrm{cont}}[t, c] = \frac{\sum_{n \in V(c)} w(n) \cdot \mathbb{1}[n \textit{ at } t]}{\sum_{n \in V(c)} w(n)}, \quad (3)$$

where $w(n) = 1/N_{\mathrm{cyc}}(n)$ is a rarity weight.

The overlap matrix O is the *seed* consumed by both Algorithm 1 (§III-D) and Algorithm 2 (§III-E).

F. Jensen–Shannon Divergence

For discrete distributions P, Q:

$$D_{\mathrm{JS}}(P \| Q) = \tfrac{1}{2} D_{\mathrm{KL}}(P \| M) + \tfrac{1}{2} D_{\mathrm{KL}}(Q \| M), \quad M = \tfrac{1}{2}(P + Q). \quad (4)$$

Bounded by $\log 2 \approx 0.693$. We measure *pitch JS* (note frequency distributions) and *transition JS*, where the latter compares *bigram distributions*—i.e., the empirical distributions of consecutive note pairs (n_t, n_{t+1}) in the original and generated sequences.

G. Greedy Forward Selection

We select a cycle subset $S \subseteq \mathcal{C}$ maximizing $f(S) = 0.5\, J(S) + 0.3\, C(S) + 0.2\, B(S)$, where J is note-pool Jaccard similarity, C is overlap-pattern Pearson correlation, and B is Betti curve similarity. Greedy selection achieves 90% preservation with $15/46$ cycles.

III. Pipeline and Generation Algorithms

A. Stage 1: Preprocessing

The MIDI file is quantized to 8th-note units ($T = 1{,}088$), two instruments are separated, and each unique (pitch, duration) pair receives a 1-indexed label ($N = 23$ notes, $C = 17$ chords for *hibari*).

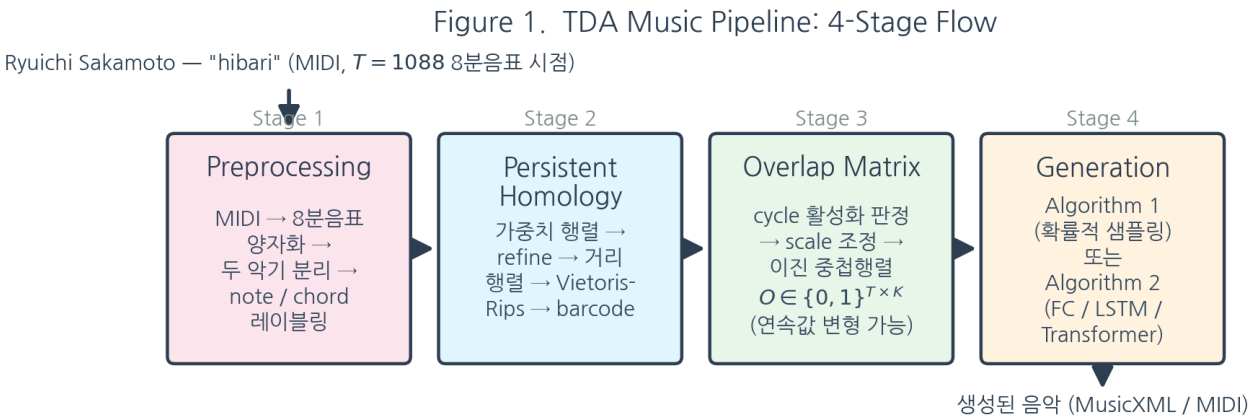

Fig. 1. Four-stage pipeline: (1) MIDI preprocessing, (2) persistent homology, (3) overlap matrix construction, (4) generation via Algorithm 1 or 2.

B. Stage 2: Persistent Homology

A weighted note-network is constructed (§II-D), a distance matrix is derived, and H_1 filtration is computed over $r \in [0, 1.5]$.

C. Stage 3: Overlap Matrix

Cycle activations are computed at every timepoint and thresholded (§II-E).

D. Algorithm 1 — Topological Sampling

For each timepoint t:

- If active cycles exist ($\sum_c O[t,c] > 0$): sample from the *intersection* $I(t) = \bigcap_{c:O[t,c]=1} V(c)$; if $I(t) = \emptyset$, fall back to the union $U(t)$.
- If no active cycle: sample from $P \setminus A(t)$, where $A(t) = \bigcup_{c:\, O[t-1,c]=1 \vee O[t+1,c]=1} V(c)$ is the union of vertices belonging to cycles active at the immediately neighboring timepoints. This prevents topological "bleeding" of cycle material into inactive regions.
- Collision avoidance: up to $R = 50$ re-samples per timepoint.

E. Algorithm 2 — Neural Sequence Models

We train three architectures on the mapping $f_\theta : O[t,:] \mapsto y[t,:]$, where y is the multi-hot note vector:

- **FC**: 2-layer fully connected ($H = 128$), timepoint-independent. A shallow architecture suffices because each timepoint's output depends only on the current overlap vector, not on temporal context.
- **LSTM**: 2-layer, forward sequential context.
- **Transformer**: 2-layer, 4-head, learned positional embeddings.

TABLE I
JS DIVERGENCE OF ALGORITHM 1 UNDER FOUR DISTANCE FUNCTIONS ($N = 20$, DECAYED LAG $\ell \in \{1, \ldots, 4\}$).

Distance	K	JS (mean $\pm$ std)	Coverage
frequency	43	0.0753 ± 0.0033	0.991
Tonnetz	**46**	$\mathbf{0.0398 \pm 0.0031}$	1.000
voice-leading	22	0.0891 ± 0.0048	1.000
DFT	20	0.0511 ± 0.0029	1.000

All models use `BCEWithLogitsLoss` with data augmentation (subset sampling, circular shift, bit-flip noise). At inference, an adaptive threshold $\theta = \text{quantile}(P, 1 - \rho_{\text{on}})$ matches the original ON ratio $\rho_{\text{on}} = |\{(t,n) : n \text{ active at } t\}|/(T \times N) \approx 0.15$, the fraction of entries that are nonzero in the original multi-hot matrix.

IV. EXPERIMENTS

All experiments use $N = 20$ independent repetitions. The primary metric is pitch JS divergence.

A. Distance Function Comparison

Tonnetz reduces JS by 47% relative to frequency (Table I). Welch's $t = 35.1$, $\nu \approx 38$, $p < 10^{-20}$, Cohen's $d = 11.1$.

B. Octave Weight Tuning

Grid search over $w_o \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ ($N = 10$) found $w_o = 0.3$ optimal (JS 0.0479, -18.8% vs. $w_o = 0.5$), consistent with *hibari*'s narrow 2-octave pitch range (MIDI 52–81).

C. Cycle Subset Ablation

JS decreases monotonically with K (Table II), but with diminishing returns: the gain from $K{:}\,17 \to 46$ (29 extra cycles) only marginally exceeds $K{:}\,10 \to 17$ (7 extra).

TABLE II
CYCLE SUBSET ABLATION (TONNETZ, $N = 20$).

K	JS (mean $\pm$ std)	Coverage
10	0.0991 ± 0.0038	0.980
17	0.0740 ± 0.0038	0.996
46	$\mathbf{0.0397 \pm 0.0025}$	1.000

TABLE III
CONTINUOUS OVERLAP EXPERIMENTS (TONNETZ, $N = 20$). DENSITY IS THE FRACTION OF NONZERO ENTRIES IN THE OVERLAP MATRIX O.

Setting	Density	JS (mean $\pm$ std)
Binary (cached)	0.751	0.0387 ± 0.0027
Continuous direct	0.264	0.0382 ± 0.0021
Cont. $\rightarrow$ bin τ=0.3	0.373	0.0386 ± 0.0022
Cont. $\rightarrow$ bin τ=0.5	**0.168**	$\mathbf{0.0343 \pm 0.0027}$
Cont. $\rightarrow$ bin τ=0.7	0.077	0.0364 ± 0.0032

D. Continuous Overlap Matrix

Continuous overlap followed by $\tau = 0.5$ binarization yields an additional 11.4% improvement over the binary baseline (Welch $t = 5.16$, $p < 0.001$, $d = 1.63$; Table III). The rarity weighting ensures rare notes contribute more to activation.

E. Deep Learning Model Comparison

The simplest FC model achieves the lowest JS (Table IV). This reflects *hibari*'s aesthetic: *out of noise* privileges spatial note placement over temporal causality, so a model that ignores time context (FC) aligns better with the compositional design.

F. Improvement F: Continuous Overlap + FC

Feeding continuous activation directly to FC as training input:

Continuous input reduces JS by 57% and val loss by $2.3\times$ (Table V). This is the lowest full-song JS observed in this study (0.09% of the theoretical max $\log 2$).

G. Song-Specific Structural Analysis

a) Deep scale property: hibari's 7 pitch classes $\{C, D, E, F, G, A, B\}$ have interval vector $[2, 5, 4, 3, 6, 1]$—all entries distinct—a property unique to diatonic scales [15].

b) Near-uniform pitch distribution: Normalized pitch entropy is 0.974 (nearly maximal), explaining FC's dominance: when all pitches are equally likely, ignoring temporal order already approximates the true distribution.

c) Tonnetz discriminability: With 7 PCs, the Tonnetz diameter is 4, providing fine-grained distance discrimination. With all 12 PCs (as in *solari*), the diameter collapses to 2, erasing Tonnetz's advantage.

d) Phase shifting: Instrument 1 repeats a 32-step module $33\times$ (no rests); Instrument 2 repeats a 33-step module $32\times$ (one rest per module). The coprime periods ($\gcd(32, 33) = 1$) ensure the two instruments never synchronize—a structure mathematically identical to Steve Reich's *phase shifting* technique.

TABLE IV
BEST DL CONFIGURATIONS (SINGLE RUN EACH).

Model	Val Loss	JS	Notes
FC (hd=256, dropout=0.3)	0.282	**0.0015**	3,754
LSTM (2-layer)	0.385	0.0448	3,753
Transformer (2L, 4H)	0.676	0.0104	3,753

TABLE V
CONTINUOUS VS. BINARY INPUT FOR FC ($N = 5$).

Input	JS (mean $\pm$ std)	Val Loss	Coverage
Binary	0.0014 ± 0.0010	0.070	0.948
Continuous	$\mathbf{0.0006 \pm 0.0005}$	**0.031**	**0.991**

V. FIGURES

Six static figures and one interactive HTML supplement are provided. Fig. 1 shows the pipeline; Fig. 2 maps *hibari*'s pitch classes on the Tonnetz lattice; Fig. 3 presents the persistence barcode; Fig. 4 compares original and generated piano rolls; Fig. 5 plots JS vs. training epoch.

VI. COMPARISON WITH RELATED WORK

A. vs. General AI Music Generation

Unlike corpus-trained models (Magenta, Music Transformer), our pipeline performs *deep analysis of a single composition*, yielding interpretable, traceable generation. The FC $>$ Transformer finding challenges the universal assumption that more temporal modeling is better.

B. vs. Existing TDA–Music Research

Compared to prior TDA–music work [6], [7]: (A) we systematically compare four distance functions with N=20 statistics; (B) we introduce continuous overlap matrices; (C) we provide full statistical rigor (Welch t, Cohen's d); (D) we extend to Western contemporary music; (E) we connect model selection to aesthetic context; (F) we extend TDA from analysis to *variation* (§VII).

VII. EXTENSION EXPERIMENTS

A. Module-Level Generation

hibari's 32-step modular structure motivates generating a single module and replicating it $33\times$. Five prototype strategies for constructing a 32-step module overlap vector were compared:

- **P0**: element-wise OR of all module instances (any-active).
- **P1**: element-wise mean across modules, binarized at τ=0.5.
- **P2**: element-wise mean without binarization (continuous prototype).
- **P3**: element-wise median across modules.
- **P5**: flat control (all entries set to the global mean).

P4 takes a different approach: instead of aggregating the full-song overlap matrix, it runs module-local persistent homology

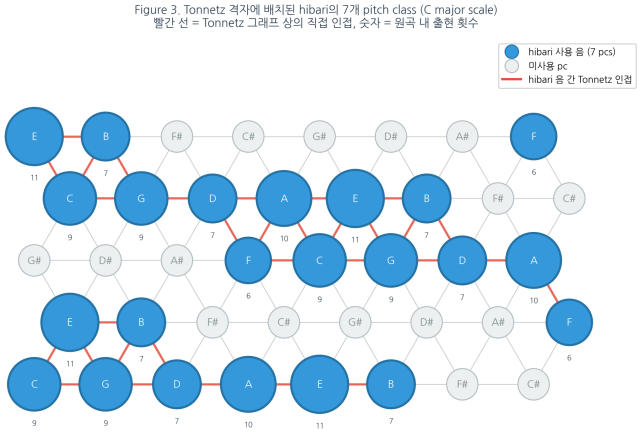

Fig. 2. Tonnetz lattice with *hibari*'s 7 pitch classes (blue). Red edges show direct Tonnetz adjacency.

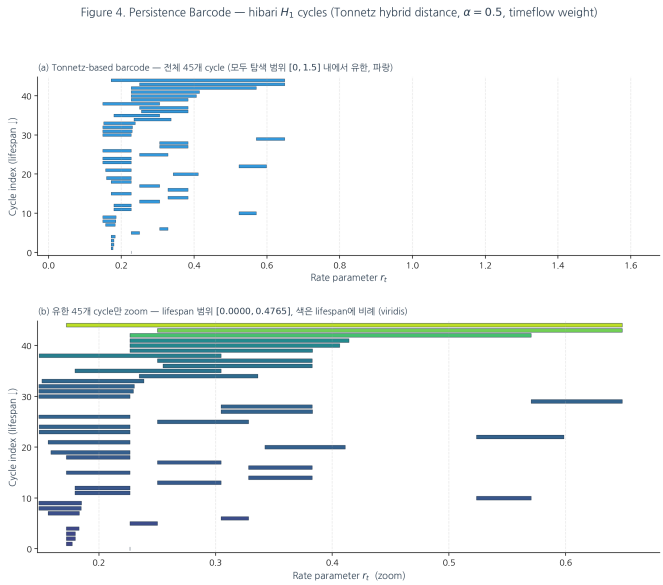

Fig. 3. H_1 persistence barcode (Tonnetz). Top: full barcode; bottom: zoomed to finite cycles. Lifespan varies by $> 10{,}000\times$.

on the first module alone (K=24 cycles). P4 combined with best-of-10 selection (C) achieved the best result:

Start-module analysis over 8 positions confirmed module 0 ($t \in [0, 32)$) as optimal (JS 0.054, best 0.026), consistent with minimalist composition practice of presenting core material in the opening module. The best single trial (seed 9302, JS $=$ 0.0258) outperforms the full-song baseline mean (0.0398).

B. Generalization: solari, *Bach, Ravel*

On *solari* (12-PC, chromatic), Transformer is optimal (JS 0.016), exactly reversing *hibari*'s FC dominance (Table VII). On Bach Fugue, Tonnetz achieves -54.8%, the largest improvement observed. On Ravel Pavane (N=49 notes), frequency is optimal, suggesting that rich note diversity favors frequency-based separation.

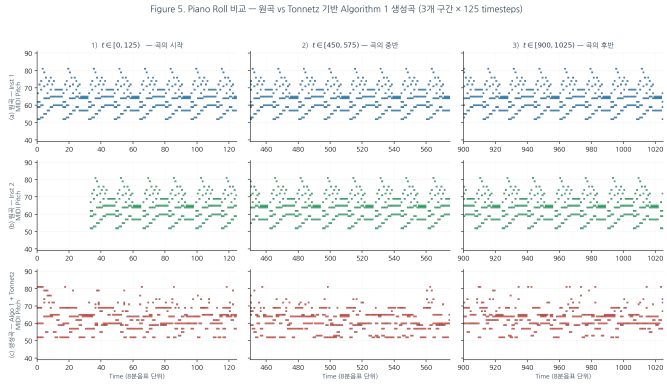

Fig. 4. Piano roll comparison: (a)–(b) original instruments 1–2; (c) Algorithm 1 + Tonnetz.

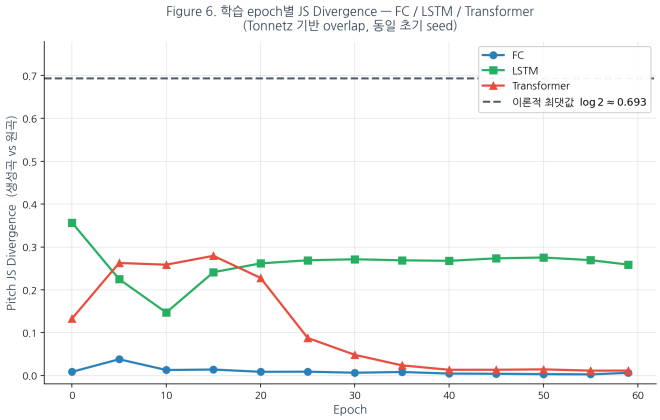

Fig. 5. JS divergence vs. training epoch. FC converges fastest; LSTM plateaus at JS ≈ 0.26.

C. Direction A: Two-Strategy Note Redistribution

The overlap matrix is preserved intact; only note identities change. We compare two exclusive matching strategies over $n_{\text{cand}} = 1000$ candidate pitch sets ($N = 20$ trials, *hibari*, Tonnetz, MIDI 48–84).

Strategy A — Positional Tonnetz nearest-neighbour (`tonnetz_nearest`):

$$\text{new_pitches}[i] = \arg\min_{p \in \text{pool}} d_{\text{Tonnetz}}\big(\text{orig_pitches}[i],\, p\big). \tag{5}$$

Each original note is replaced by its closest pool element in Tonnetz space; the index i encodes a 1:1 positional correspondence that preserves local harmonic relations.

Strategy B — Matrix-structure Hungarian matching (`ascending`):

$$\pi^* = \arg\min_{\pi} \left\| P_\pi \, \hat{D}_{\text{new}} \, P_\pi^\top - \hat{D}_{\text{orig}} \right\|_F, \tag{6}$$

where $\hat{D}$ is the off-diagonal min-max normalised distance matrix:

$$\hat{D}_{ij} = \frac{D_{ij} - \min_{k \neq l} D_{kl}}{\max D - \min_{k \neq l} D_{kl}}, \quad \hat{D}_{ii} = 0. \tag{7}$$

TABLE VI
MODULE-LEVEL GENERATION STRATEGIES ($N = 10$).

Strategy	JS (mean±std)	Best	Cov.
P1 baseline	0.114 ± 0.039	0.074	0.77
C: best-of-10	0.074 ± 0.027	0.024	0.90
D: cov≥20/23	0.082 ± 0.022	0.050	0.88
P4: local PH	0.066 ± 0.019	0.038	0.84
P4+C	$\mathbf{0.059 \pm 0.015}$	**0.026**	0.89

TABLE VII
CROSS-COMPOSITION SUMMARY. "—" INDICATES THAT ALGORITHM 2 (DEEP LEARNING) WAS NOT TESTED FOR THAT PIECE; ONLY ALGORITHM 1 RESULTS ARE REPORTED.

Piece	PCs	Best dist.	Best model	Interpretation
hibari	7	Tonnetz	FC	Spatial placement
solari	12	freq/Ton.	Transformer	Melodic progression
aqua	12	Tonnetz	—	Tonnetz $+26.3\%$
Bach	12	Tonnetz	—	Tonnetz -54.8%
Ravel	12	frequency	FC	Rich distribution

Solving the joint row-column permutation exactly is NP-hard; with $N = 17$ notes, $N! = 3.56 \times 10^{14}$ exhaustive evaluations are infeasible, so all experiments use the Hungarian approximation (`scipy.optimize.linear_sum_assignment`).

Both strategies report **note_dist_error** $= \|\hat{D}_{\text{orig}} - \hat{D}_{\text{new}}\|_F$, which is on the same $\hat{D}$ scale and directly comparable across rows. Applying both strategies simultaneously degrades performance (Strategy A's positional semantics are overwritten by Strategy B's permutation), so the two are applied exclusively via `matching_mode`.

Strategy A achieves pitch JS 0.2930 vs. 0.3815 for Strategy B (-23.2%, Table VIII). Local pairwise harmonic preservation (A) outperforms global distance-matrix alignment (B), consistent with *hibari*'s small note vocabulary ($N = 17$) in which adjacent Tonnetz relations directly govern harmonic character (§IV-G). The wider pitch range vwide (MIDI 40–88) was inferior to wide (48–84) in all conditions; wide is adopted.

D. Direction B: Temporal Reordering

Two strategies were tested: block permutation (dividing the sequence into fixed-size blocks and randomly permuting them; also called "segment shuffle" in our codebase) and Markov resampling. With PE removal and retraining, Transformer achieves DTW $+21.7\%$, but pitch JS degrades from 0.007 to 0.173. Direction B alone cannot simultaneously preserve pitch distribution and alter melody; it requires combination with Direction A.

E. Harmonic Constraints

Scale constraints (major, minor, pentatonic) and consonance objectives were added to note redistribution. **scale_major** was most effective: note error $4.35 \to 3.52$ (-19%), scale match 1.00, average dissonance $0.412 \to 0.361$.

TABLE VIII
TWO-STRATEGY COMPARISON (ALGORITHM 1, TONNETZ, MIDI 48–84, $N = 20$, $n_{\text{cand}} = 1000$).

Strategy	pitch JS	trans. JS	DTW	note err
baseline (original)	0.0212	0.2470	2.2186	—
B: ascending (Hungarian)	0.3815	0.6121	2.7918	4.6819
A: `tonnetz_nearest`	**0.2930**	**0.5488**	2.6442	**3.2132**

TABLE IX
PER-CYCLE τ_c OPTIMIZATION ($N = 20$).

Setting	JS (mean ± std)	Improvement
Uniform $\tau = 0.35$	0.0460 ± 0.0034	—
Per-cycle τ_c	$\mathbf{0.0241 \pm 0.0023}$	$\mathbf{+47.5\%}$

F. Combined A+B: Final Integration

The best combined setting (**major_block32**): scale_major redistribution + block permutation(32) + continuous overlap + Transformer achieves: ref pJS $= 0.002$ (near-perfect learning), DTW $+31\%$ (different melody), C major scale match $= 1.00$.

G. Per-Cycle Threshold Optimization

Greedy coordinate descent over $\tau_c \in \{0.1, 0.2, \ldots, 0.7\}$ for each cycle:

47.5% improvement ($p < 0.001$; Table IX). The majority of cycles prefer $\tau < 0.35$, suggesting that "accompaniment-type" cycles were previously suppressed. No overfitting: N=5 greedy result (0.0238) matches N=20 (0.0241).

H. Soft Activation: All Architectures

FC and Transformer show massive improvement with continuous input; LSTM is incompatible (Table X). FC + continuous achieves JS $= 0.0004$—the overall best in this study.

I. Tonnetz α Grid Search

$\alpha = 0.0$ (pure Tonnetz) is individually optimal, but combined with $w_o = 0.3$ and decayed lag, cycle count collapses to $K = 14$. To maintain topological diversity ($K = 42$), we retain $\alpha = 0.5$ as the default (Table XI).

J. Barcode Wasserstein Module Selection

For each candidate start module at position m, we compute the persistence diagram PD_m from the module-local chord sequence (32 timepoints, single instrument) and compare it to the full-song persistence diagram PD_{full} via the Wasserstein distance $W_1(\text{PD}_m, \text{PD}_{\text{full}})$. The hypothesis is that modules whose local topology most closely resembles the global topology will yield lower generation JS.

Across 8 start positions, the Pearson correlation between W_1 and generation JS is $r = 0.503$—moderate but not decisive. Three caveats limit its applicability:

1) **Chord-space mismatch**: module-local PD is computed from a 32-step window, whose chord vocabulary is a strict subset of the full song's. The resulting PD inhabits a smaller simplicial complex.

TABLE X
BINARY VS. CONTINUOUS INPUT ($N = 5$).

Model	Input	JS	Val Loss	ΔJS
FC	Binary	0.0035	0.307	—
FC	**Cont.**	**0.0004**	**0.026**	**+88.6%**
Trans.	Binary	0.0034	0.465	—
Trans.	**Cont.**	**0.0007**	**0.121**	**+79.4%**
LSTM	Binary	0.280	0.411	—
LSTM	Cont.	0.290	0.411	−3.5%

TABLE XI
α GRID SEARCH ($N = 20$).

α	K	JS (mean $\pm$ std)	Note
0.0	14	$\mathbf{0.0574 \pm 0.0023}$	Best among $K \geq 14$
0.3	48	0.0870 ± 0.0046	Worst
0.5	42	0.0594 ± 0.0033	Default
1.0	1	0.0351 ± 0.0020	Degenerate (K=1)

2) **Single-instrument limitation**: module-local PH uses only one instrument's note sequence, so inter-instrument weight information (which contributes the largest JS gains via decayed lag weighting) is entirely absent.
3) **Scale sensitivity**: W_1 is sensitive to the absolute scale of birth/death coordinates, which differ between a 32-step and a 1,088-step filtration.

We therefore position Wasserstein module selection as a supplementary tiebreaker rather than a primary selection criterion.

VIII. CONCLUSION

We presented a unified pipeline that extracts topological structure from music via persistent homology and generates new music preserving that structure. Key findings:

1) **Distance function matters.** Tonnetz reduces JS by 47% on *hibari* ($p < 10^{-20}$), but this advantage is composition-specific: Bach Fugue shows -54.8% Tonnetz improvement, while Ravel Pavane favors frequency.
2) **Composition character determines optimal model.** *hibari* (diatonic, entropy 0.974) $\rightarrow$ FC; *solari* (chromatic, entropy 0.905) $\rightarrow$ Transformer.
3) **Continuous overlap is powerful.** Per-cycle thresholding adds 47.5% ($p < 0.001$); soft activation input yields $+88.6\%$ (FC) and $+79.4\%$ (Transformer).
4) **Topology-preserving variation is feasible.** Note redistribution + temporal reordering + harmonic constraints produce music with DTW $+31\%$ (different melody), ref pJS 0.002 (accurate learning), and perfect C major scale match.
5) **Module-level generation is competitive.** P4+C achieves JS 0.059 (average) with best trial 0.026 outperforming the full-song baseline—extending the pipeline to real-time interactive composition tools.

This work demonstrates that TDA can serve not only as an analysis tool but as a *constraint generator for musical creation*, bridging mathematical topology and compositional aesthetics.

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